

ORIGINAL ARTICLE

Explainable classification by local error amplification

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Abstract

In this study, we introduce a novel predictive approach rooted in the field of explainable AI, designed to provide sketches of the action of a non-interpretable machine learning model. The use case is psychological profiling and concerns the alignment between individuals' personalities and specific job roles within the fields of psychology and career development. Our approach builds upon the foundation of BAPC (Before and After prediction Parameter Correction) (Sobieczky and Geiß in Explainable AI by BAPC—Before and After correction Parameter Comparison, 2023), an explainable AI technique designed to elucidate predictions from models with interpretable parameters of a surrogate 'base' model. We extend this method by incorporating error amplification (EA), a new technique effectively integrating decision trees as our base model. In the three-step methodology, rooted in the principles of BAPC, enhanced by error amplification, we firstly train a base model using decision trees and identify the model's errors. Second, we introduce an AI model to predict the errors made by the base model. Subsequently, we amplify these predicted errors in a localized manner around the individual point of interest. In the third step, we retrain the base model using the error-amplified data set. The disparity between the decision boundaries generated by the initial and the retrained decision trees provides explainable insights into enhancing specific personality features. While our method finds its application in personality-job alignment here, its versatility extends to a wide range of scenarios, offering valuable insights into data set feature interpretability by elucidating their effective changes.

Keywords Explainable AI · Local surrogate model · Decision support by explainable AI · Feature importance

1 Introduction

1.1 Problem formulation

Classification tasks with complicated decision boundaries are typically approached by modern machine learning methods [2], specifically by deep learning methods [3] if the data reservoir is sufficiently large. At the same time, these artificial intelligence (AI) models usually lack the ability to interpret model parameters after training [4].

Explainable AI (XAI) sets out to provide explanations that make a predictive model's output interpretable in terms of its input features [5, 6]. In the case of local surrogate models, these explanations are further enriched with context-specific information about the input space in the neighborhood of a given instance. This local context can support decision-making [7] and enable users to understand how to potentially control the model's outcome [8].

For AI-decisions based on data from personal profiles it is particularly important to provide transparency of the decision process which XAI delivers in terms of evaluating the importance of features leading to a particular decision [9, 10]. In this paper, we focus on delivering this kind of transparency in a post-hoc approach in which



the explanations are delivered independently of the learning model used for the prediction: By using a hybrid model consisting of a decision tree and one further correcting, but non-interpretable AI model for prediction - and locally reproducing this hybrid model's prediction with the interpretable surrogate model alone - given by the decision tree. The shift in the decision boundary as a (local) interpretation of the action of the AI-correction is - to the best of our knowledge - a novel contribution to XAI-literature.

1.2 Related work

Explainable AI has seen a strong rise in interest during the last decade, strongly boosted by the DARPA-initiative [11], as shown in numerous review articles [5, 6, 12].

Post-hoc explainability approaches (as opposed to intrinsic ones) refer to methods that take an already trained black-box model as input and generate explanations for its predictions as output. In this context, the term of explainability necessarily takes an independent view of the action of the model to be explained. This perspective can be crucial in fairness-related issues, as 'case-based explanations have less impact on fairness-criteria' [13]. All metrics evaluating the performance of rendering an explanation have to be defined without reference to the model itself. Furthermore, it is common practice to categorize all XAI approaches into global and local. This distinction refers to one (single) explaining model either to address the prediction of any arbitrary incidence's prediction, or to address only selected ones inside a restricted sub-area of the input space. The latter, the so-called local approaches typically use the neighborhood of given instances within which the explanations 'count'. Mostly, not much is known about the size of these neighborhoods for the explaining model to be exact (to the predictions, i.e., to have high fidelity), which is referred to as the locality problem. Furthermore, the robustness of local models under changes of the considered neighborhood must be strongly questioned [14]. When it comes to designing an interactive aid for decision making, the full application involves a neural-symbolic cycle (see Sect. of [14]) which forces "one-off descriptions" to be reconsidered for further corrections. Aiming at the consistency between corrected model and explained AI-modeling in the *local* category of AI-explanations even more needs the control of fidelity, as otherwise mistakes can built up in the repetition of the corrections. As our method provides control of the local horizon of the surrogates applicability, we claim to give an answer to this request of Hudec et. al.. Furthermore, as noted in [15] (see Sect. 2.1.3), the local surrogates don't "compare the level of alignment with the expert's explanation." This discrepancy even stronger calls for control of the range of validity of explanations derived from local surrogates to keep the contribution of the explanation derived from the XAI-method clearly defined and potentially distinct from the expert's explanation.

Local surrogate modeling (LSM) is an important subfield of Explainable AI. LSM is a post-hoc, model-agnostic XAI-method [16] which aims at finding an appropriate representative set of neighboring instances which allow the evaluation of the prediction in the light of a local trend.

Apart from SHAP [17], and ANCHORS [18], the original LIME concept [19] is the most appropriate to compare our approach of Error Amplification (EA) to, as it also modifies the original data set: It also uses the induced changes to derive explanations but instead of locally sampling synthetic data (LIME) it changes the labels of at existing instances (EA). SHAP is different in that it resamples the given data set without changing it. As we point out the improvement of the local surrogate idea *with* changing the data-set, we focus on showing the advantages compared to LIME. In particular, the strong dispersion in the results of the interpretations is shown to vanish when using BAPC-DT. LIME has seen strong improvements [20–22]. In particular, the stability problem associated to the random sampling process Section 4.4.3 in [6], and the locality problem [23] (i.e., the size of the neighborhood in the input space of the incidence for which the classification is to be explained) is subject to current research [24]. However, the optimal choice of this neighborhood is critical (see discussion in [25]), and is also subject of our results. The improvements of LIME as well as ANCHORS [18] mentioned above do not give a rule for the optimal neighborhood. Therefore, using the original version is sufficient for the comparison with our results.

1.3 Original BAPC framework

Our method extends the BAPC (Before and After Prediction Parameter Correction) framework introduced by Sobieczky (2023) [1], which is an explainable AI approach aimed at interpreting model predictions through the parameters of a transparent surrogate or “base” model. In BAPC, it is assumed that a base model with interpretable parameters exists. An AI model then provides correction terms that enhance the predictive performance of this base model. These corrections are modeled by a surrogate, which captures the differences between the original labeled data and a modified version where the labels are replaced by the AI-enhanced predictions. This setup allows for a comparison between using the base model on raw data versus applying it after incorporating the AI-driven corrections. The resulting changes in the base model’s behavior can be interpreted as effective shifts in its parameters, offering insight into how the AI model adjusts the prediction process.

While the original BAPC framework is designed for regression tasks, our work extends it to accommodate decision trees as base models.

2 Definition of the method

2.1 Problem formulation

We are considering discrete data points, $\mathcal{D} = \{\langle X_i, Y_i \rangle\}_{i=1}^I$, with $X_i \in \mathbb{R}^p$, $Y_i \in \{0, 1\}$, with p number of features in the given data set, and Y_i labels of the data points. We can summarize the method in three steps.

- (1) **Step 1.** In the **first step** we predict the labels of our data set with a simple base model, using a decision tree,

$$f : \mathbb{R}^p \rightarrow \{0, 1\}. \quad (\text{DT}_1)$$

The decision tree predicts the true labels Y_i with some prediction error ε_i ,

$$\begin{aligned} Y_i &= f(X_i) + \varepsilon_i, \quad i \in \{1, \dots, I\} \\ \text{with } \varepsilon &= \{\langle x_i, \varepsilon_i \rangle\}_{i=1}^I \end{aligned} \quad (1)$$

with a set of predicted decision boundaries.

- (2) **Step 2.** In the **second step**, we train an AI model,

$$\mathcal{A} : \mathbb{R}^p \rightarrow \{-1, 0, 1\} \quad (2)$$

on the prediction error ε_i of the first decision tree. The trained AI model predicts the error correction $\hat{\varepsilon}_i$. The overall error can be described as the sum of the prediction of the AI model and a remaining error term,

$$\varepsilon_i = \underbrace{\mathcal{A}(X_i)}_{=\hat{\varepsilon}_i} + \Delta\varepsilon_i. \quad (3)$$

- (3) **Error Amplification.** We amplify the predicted error with relabeling of neighboring data points, in order to allow and amplify changes of the decision boundaries within a prediction of a second decision tree. For the error amplification we focus on one data point of interest X_n , which typically is a data point with an error $\varepsilon_n = \{-1, 1\}$. In a circular neighborhood $\mathcal{B}_r^n(X_i)$, with the radius r and center point X_n , we relabel all data points to Y_n .

This relabeling is guided by the construction of an error-amplified vector ε_{EA} . Specifically, for data points

$X_i \in \mathcal{B}_r(X_i)$ that are correctly predicted (i.e., with $\hat{\varepsilon}_i = 0$), the error value is set to the negative of the error observed at X_n .

$$\hat{\varepsilon}_{EA} := \begin{cases} -\hat{\varepsilon}_n & \forall X_i \in \mathcal{B}_r(X_i) & \& \quad \hat{\varepsilon}_i \stackrel{!}{=} 0 \\ 0 & \forall X_i & \text{with } \hat{\varepsilon} \neq 0 \end{cases}$$

The modified relabeled dataset is then $Y' = Y + \hat{\varepsilon}_{EA}$. This leads to the modified data set $\mathcal{D}' = \{(X_i, Y'_i)\}_{i=1}^I$. The error amplification and relabeling procedure described here is specifically designed for binary classification tasks, where labels take values in $\{0, 1\}$. Consequently, label flips correspond to switching between these two classes only. Therefore, the relabeled and modified data set $\mathcal{D}'$ remains within the binary label space $\{0, 1\}$.

Extending the approach to multi-class classification represents a valuable direction for future research. This would involve generalizing the error term ε_n to take values in $\mathbb{Z}_{+,-}$, and adapting the error amplification routine to handle a wider range of prediction outcomes - e.g., where class 1 might be misclassified as class 3 or 4. Such an extension would also require developing a principled strategy to interpret and manage misclassifications that involve larger gaps between class labels.

To find the best radius r , we calculate the R^2 score (see 2.1.2) for all data points in the circle and select the radius r with a maximal R^2 score.

- (4) **Step 3.** In the **third step** we use this modified data set $\mathcal{D}'$ as input to train a second decision tree, which leads to a new prediction of the data labels and to new decision boundaries. In comparing the decision boundaries of the first decision tree to those of the second, we can identify which features need to be adjusted to achieve a desired alteration in the categories for a specific data point of interest. By defining Feature Importance, we can determine which features are most relevant in influencing changes in class labels.

Algorithm 1: BAPC-DT with Error Amplification

BAPC-DT with Error Amplification

Input-1 : Discrete labeled data points $\mathcal{D} = \{\langle X_i, Y_i \rangle\}_{i=1}^I$
with $X_i \in \mathbb{R}^p$, $Y_i \in \{0, 1\}$,

Input-2 : Data point of interest $X_n \in X_i$

Output-1: Feature Importance - Change in decision boundaries of first and second decision tree.

Output-2: Fidelity measure ϕ

Output-3: Surrogate model prediction $\hat{Y}_i^{surr} = f(X_i) + \mathcal{A}(X_i)$

Initialize

1	Instantiate f_{X_i}	// instantiate base decision tree (DT)
		// model
2	Instantiate $\mathcal{A}(X_i)$	// instantiate error prediction AI model
3	Instantiate $r \in \mathbb{R}^+$	// instantiate neighborhood radius for
		// error amplification
	Step-1	// First application of the base DT-model
4	$\tilde{Y}_i \leftarrow \text{FitAndPredictWithDT}(f_{X_i}, Y_i)$	// fit and predict
5	$\varepsilon_i = \tilde{Y}_i - Y_i$	// residuals
	Step-2	// Application of the error prediction AI model
6	$\hat{\varepsilon}_i \leftarrow \text{FitAndPredictWithAI}(\varepsilon)$	// fit and predict
7	$\hat{\varepsilon}_{EA} = \begin{cases} -\hat{\varepsilon}_n & \forall X_i \in \mathcal{B}_r(X_i) \text{ \& } \hat{\varepsilon}_i = 0 \\ 0 & \forall X_i \text{ with } \hat{\varepsilon} \neq 0 \end{cases}$	// error amplification
8	$Y'_i = Y_i + \hat{\varepsilon}_{EA}$	// relabeled dataset
	Step-3	// Second application of the base DT-model
9	$\tilde{Y}'_i \leftarrow \text{FitAndPredictWithDT}(f_{X_i}, Y'_i)$	// fit and predict
10	$\phi \leftarrow 1 - \frac{(\tilde{Y}' - \tilde{Y}) - \hat{\varepsilon}}{n_{neighborhood}}$	// fidelity measure
11	$FI \leftarrow 1/\text{min_distance}(\text{decision_boundary}, X_n)$	// feature importance

2.1.1 Feature importance

Most of the explainable AI methods provide a so called feature importance analysis: For local surrogates this amounts to a kind of sensitivity analysis. However, instead of computing gradients, which effectively render the problem a regression problem, the feature importance in the case of BAPC for classification is defined as the reciprocal univariate distance to the respective nearest decision boundary:

For each feature direction, we calculate the minimal distance from the data point X_n and the decision boundary of the second decision tree.

$$FeatureImportance_j = \frac{1}{\min\{|X_n(j) - db_j|\} / \max(x(j))}. \quad (4)$$

with $X_n(j)$ being the j -coordinate of the data point of interest X_n and db_j being a decision boundary in j -feature direction. The distance is normalized with the maximal value of available data points in feature direction j , $\max(x(j))$.

This provides a measure of how much influence the feature has on the decision, offering insights into which features are most critical for classification. If a data point is close to a decision boundary, it means the feature is significant. Even a small change to that feature could result in a different predicted class label.

2.1.2 Fidelity

Fidelity measures how accurately an explanation reflects the predictions of a black-box model. It is defined as the difference between the explanation and the correction applied by the AI model, denoted as $\hat{\epsilon}$:

$$\phi = (\tilde{Y}' - \tilde{Y}) - \hat{\epsilon}. \quad (5)$$

Fidelity quantifies how accurately an explanation reflects the predictions of a black-box model. So in other words, we want to evaluate if the explanation that we are giving actually represents what the black-box model is doing. In the context of BAPC with Error Amplification, we use two complementary approaches to evaluate fidelity.

The first approach compares the actual change in predictions of the base model - before and after error amplification - with the predicted correction from the AI model. Specifically, we compare $(\tilde{Y}' - \tilde{Y})$ to the predicted error $\hat{\epsilon}$. This leads to the general fidelity formulation:

$$(\tilde{Y}' - \tilde{Y}) - \hat{\epsilon}, \quad (6)$$

where $\hat{\epsilon}$ takes values in the discrete set $\{-1, 0, 1\}$. To evaluate fidelity across a neighborhood of the point of interest X_n , we use an aggregated measure:

$$\phi = 1 - \frac{\sum_i (\tilde{Y}'_i - \tilde{Y}_i) - \hat{\epsilon}}{n_{neighb_points}}, \quad (7)$$

where the sum runs over all data points i in the circular neighborhood $\mathcal{Br}(X_n)$, and n_{neighb_points} is the number of points in that neighborhood. The subtraction from 1 in Eq. (7) transforms the average discrepancy into a fidelity score, such that higher values indicate stronger agreement between the explanation and the predicted correction.

In addition to this direct comparison, we also evaluate fidelity using the coefficient of determination (R^2), a standard regression metric:

$$R^2 = 1 - \frac{SS_{res}}{SS_{tot}}, \quad (8)$$

$$SS_{res} = \sum (\hat{\epsilon} - (\tilde{Y}' - \tilde{Y}))^2 \quad (9)$$

$$SS_{tot} = \sum (\hat{\epsilon} - \text{round}(\text{mean}(\hat{\epsilon})))^2. \quad (10)$$

The R^2 -score was chosen for comparability with LIME, which also uses R^2 as its fidelity metric.

The calculation of the coefficient of determination involves comparing the prediction errors of the AI model, $\hat{\epsilon}$, with the differences between the predictions of the second decision tree, $\tilde{Y}'$, and the first decision tree, $\tilde{Y}$.

When computing SS_{tot} , we account for the discrete nature of classification labels by rounding the mean of $\hat{\epsilon}$. Specifically, this involves measuring the difference between each $\hat{\epsilon}$ value and the most frequent value in $\hat{\epsilon}$, allowing us to analyze deviations from the most common value.

We compute the R^2 score for all data points within the circular neighborhood of the data point of interest, X_n . A high R^2 value (close to 1) indicates a high degree of fidelity and reliable local explanations. In contrast, a low R^2 value (close to 0 or negative) indicates that the explanation poorly corresponds to the correction made by the AI model. This misalignment may result from limitations and the simplicity of decision trees, poor predictions of the AI model or suboptimal parameter selection, like the neighborhood radius for Error Amplification.

2.1.3 The explanation is the difference in decision boundaries

BAPC with Error Amplification (EA) provides an explanation of how features can be adjusted to trigger changes in class label predictions. When we update the model with new data, where several labels of nearby instances of the data point of interest X_n are changed to the same predicted label Y_n , the new decision tree ($DT2$) results in updated decision boundaries. As a result, the input space is divided into two distinct regions, each corresponding to one of the decision trees.

From the perspective of probability theory, these two regions represent “events.” By the change in the decision boundary there is a change in the partition of the input space, and thus a change in the corresponding events. Looking at all possible intersections of events from either partition, we get sub-regions of the input space within which the same classification of the two decision trees results and which allows a more refined distinction between the two fitted models. This can be expressed in the language of σ -algebras, in which the refinement would be described by the joint σ -algebra of the two decision tree models.

3 Applications: performance evaluation and explainability analysis

In this section, we demonstrate the applicability and effectiveness of our proposed method, BAPC with Error Amplification, through two types of datasets: synthetic data and real-world data reflecting personality features. The synthetic dataset provides a controlled environment to evaluate the performance and explainability of the method, while the real-world dataset showcases its practical relevance. For both cases, we benchmark our approach against an established local surrogate model, LIME [19], offering a comparative analysis of performance and explainability. This dual perspective highlights the strengths of BAPC in delivering insightful explanations for AI-supported predictions.

3.1 Synthetic data case study: performance and explainability

BAPC with Error Amplification is demonstrated on a simulated dataset consisting of two features, $\{\text{feat}_1, \text{feat}_2\} \in [0, 7]$, and binary class labels $Y_i \in \{0, 1\}$. The class labels are determined using a hyperbolic decision boundary defined by the formula:

$$Y_i = \begin{cases} 1 & \forall \quad X_i < \frac{5}{(X_i - 0.5)} + 0.5 \\ 0 & \text{else.} \end{cases}$$

In this example, we focus on a specific data point, $X_n = [2.5, 2.6]$, which has a true class label of $Y_n = 1$. Our goal is to understand why this data point is predicted to belong to its current class, and to identify which features would need to change, and in which direction, in order for it to be classified into a different class. The specification parameters which were used for this simulation example are summarized in Table 1.

In the first step of BAPC with Error Amplification (Fig. 1a) we predict the labels Y of the data set with the first base decision tree model, which yields to the class predictions $\tilde{Y}$. In this first prediction the data point of interest X_n is predicted to belong to the class category $\tilde{Y}_n = 0$, which is a prediction error.

In the second step (Fig. 1b) an AI model, here the Random Forest Classifier is trained with the error which is made by the first decision tree $\varepsilon = (Y - \tilde{Y})$ and gives a prediction of the error $\hat{\varepsilon}$. The combination of the trained base decision tree and the trained AI model builds the surrogate model and can then be used for the label prediction of new data points (e.g., for test data).

Table 1 Specification parameters for running BAPC with Error Amplification on the synthetic data set

Parameter	Value	Information
$n_samples$	1000	Total number of data points $\{X_i\}_{i=1}^{n_samples}$
X_n	[2.5, 2.6]	Data point of interest
max_depth	2	Maximum depth of the decision tree
min_samples_leaf	2	Minimum number of samples required to be at a leaf node
min_samples_split	2	Minimum number of samples required to split an internal node
cost_fct-DT	Relative Accuracy Measure	Cost function for decision tree to measure the quality of a split
AI-model	Random Forest Classifier	Which AI-model to be used

Now, the error is amplified by relabeling data points in the circular neighborhood of the data point of interest X_n , within a circle $\mathcal{B}_r(X_i)$ with radius r (Fig. 1c). Hence, data points within this circle, that were actually predicted correctly ($X_i \in \mathcal{B}_r(X_i)$ with $\hat{\epsilon}_i = 0$), are now assigned the negative error value of the error value of the data point of interest X_n ,

$$\hat{\epsilon}_{EA} := \begin{cases} -\hat{\epsilon}_n & \forall X_i \in \mathcal{B}_r(X_i) & \& \hat{\epsilon}_i \stackrel{!}{=} 0 \\ 0 & \forall X_i & \text{with } \hat{\epsilon} \neq 0, \end{cases}$$

with $\hat{\epsilon}_{EA}$ being the error amplified vector (Fig. 1c).

In step three, we modify the labels of the original data set while adding the error amplified $\hat{\epsilon}_{EA}$ vector to the original data labels, $Y' = Y + \hat{\epsilon}_{EA}$ (Fig. 1d).

Next, a second decision tree model, identical in properties to the one used in the first step, is trained on the newly modified dataset. This results in a shift of the decision boundaries, which is depicted by the green arrows and the green decision boundaries in Fig. 1d, and thus in a new prediction of the labels $\tilde{Y}'$. These shifts in the decision boundaries reveal which features of the data point X_n could be adjusted to change its classification from 1 to 0.

In this case, our method suggests that *feature2* has a higher importance, compared to *feature1* since the data point X_n is much closer to the decision boundary in *feature2* direction, so changing its values in this direction would result in a change in class label prediction. The R^2 score in this example is 0.44.

3.2 Ablation study - BAPC without Error Amplification

To illustrate the importance of the newly integrated error amplification step in the BAPC framework, we perform an ablation study by applying the method to the synthetic dataset while omitting the error amplification step (Appendix: Fig. 5). The initial steps of the method remain unchanged: in Step 1 a decision tree is trained on the original training dataset (Fig. 5a). In the Step 2, we identify the error which is made by the decision tree prediction ($Y - \tilde{Y}$) and predict the errors with an AI model. Here, we focus on the errors in the neighborhood of the data point of interest (Fig. 5b).

In the third step, we modify the training dataset by subtracting the raw predictions $\hat{\epsilon}$ (i.e., without applying error amplification) directly from the original labels, resulting in $Y - \hat{\epsilon}$ (Fig. 5c). This operation relabels the few misclassified class 1 data points – those located within the class 0 decision region – as class 0. As a result, the entire decision region becomes uniformly labeled as class 0. As a result, when we train a second decision tree on this modified dataset, it *learns the same decision boundaries as the first tree*, because there is not enough incentive to change them. This happens because the relabeling reinforces the original class 0 regions even more

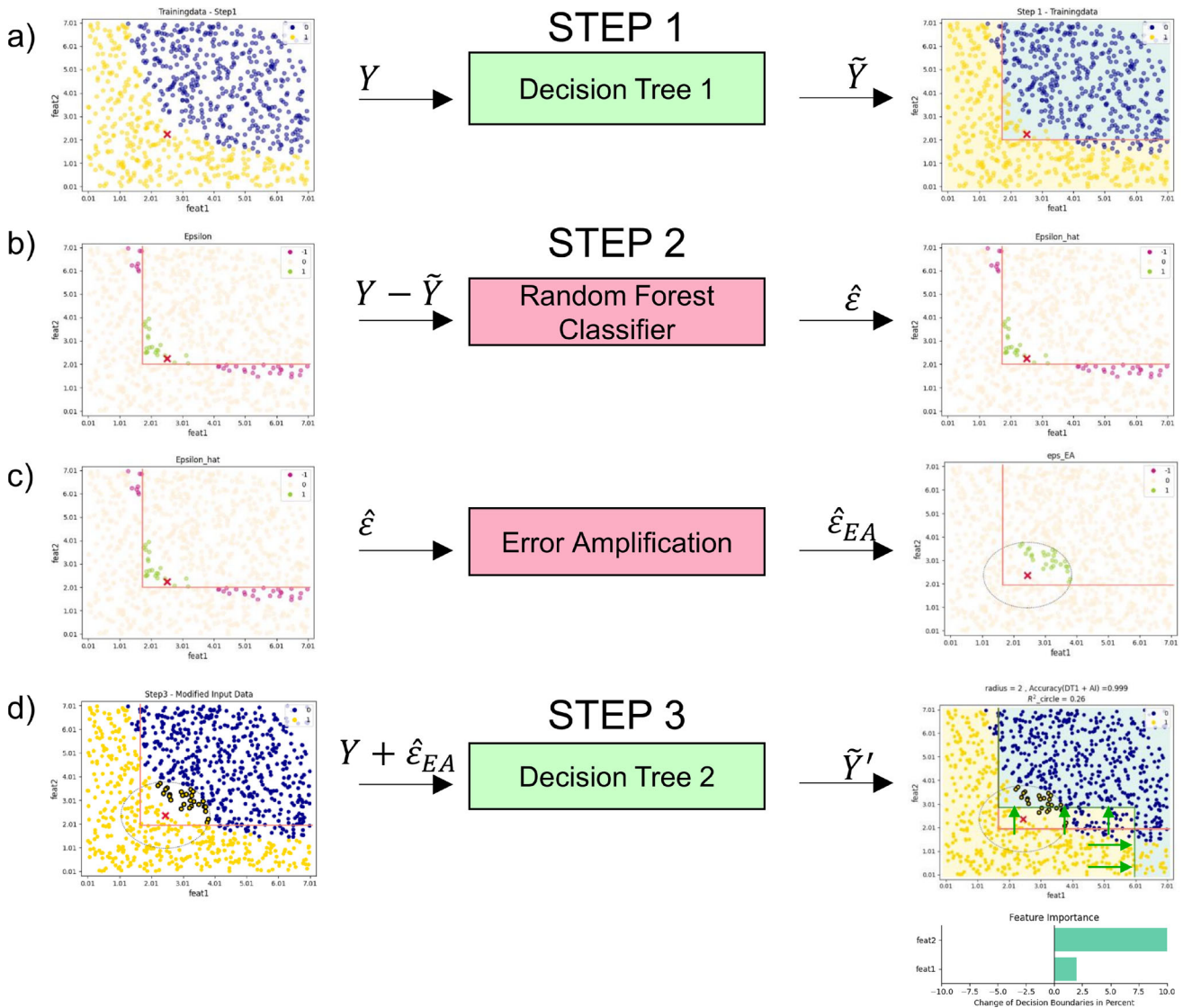


Fig. 1 Schematic Overview of BAPC-DT with Error Amplification. Discrete data points are shown for two features belonging to two categories $\{0\text{--blue}, 1\text{--yellow}\}$. The data point of interest X_n is depicted with the red cross. The red/green lines correspond to the decision boundaries before/after error amplification. **a** Step 1: Train first decision tree model and get predictions $\tilde{Y}$. **b** Step 2: Infer the error of the predictions of the first decision tree $Y - \tilde{Y}$ and predict the error with an AI model $\hat{\epsilon}$. **c** Error Amplification: The error is amplified while relabeling data points within a circular neighborhood of the data point of interest X_n . **d** Step 3: A second decision tree is trained on the modified data set which leads to a shift of the decision boundaries. The importance of the features are depicted in the Feature Importance bar plot (color figure online)

strongly. Shifting the decision boundaries in this case would only introduce more classification error, which the decision tree is designed to avoid.

To truly probe the model's behavior and encourage changes in the decision boundaries (which helps us understand which features drive the decisions and in which direction), we need to amplify the error signal. This increases the number of class 1-labeled points within the original classified class 0 regions, giving the decision tree a reason to adjust its boundaries. That is why the error amplification step is essential for meaningful feature importance analysis.

3.3 Case study on application data: performance and interpretability

In this section, we apply our BAPC method with Error Amplification to a real-world dataset provided by an anonymous industry partner from the consulting and personal services sector.

This dataset evaluates individual performance based on various critical personal traits that are essential for success in leadership roles. It has 372 instances with 10 features representing personality traits measured on a discrete scale with seven levels, extracted from survey data.

The objective of this study is to offer individuals actionable insights into their assessed personal attributes, identifying which traits they can strengthen or develop further to enhance their suitability for leadership positions.

The dataset consists of evaluations for each individual across ten key personal trait features. Using domain expertise, these feature distributions were analyzed and labeled to determine whether an individual's trait profile indicates suitability for a leadership role (class label = 1) or not (class label = 0). The labeling process is grounded in domain knowledge, supported by statistical insights drawn from previous evaluations. This domain knowledge is itself informed by prior studies demonstrating correlations between specific personality traits and leadership effectiveness [26–28]

Our aim is to train a predictive model on this dataset that not only assesses new individuals' potential for leadership positions but also provides clear explanations of which traits could be improved to increase their likelihood of qualifying for such roles.

As a case study, we focus on an individual referred to as person #244, who is interested in understanding whether adjustments to certain personality traits could improve their suitability for a leadership position. To explore this, we apply the BAPC method with Error Amplification to examine shifts in decision boundaries across various feature pairs (Fig. 2).

Specifically, we analyze the feature pair “Willingness to Compromise” and “Activity Level” (Fig. 2a). The analysis reveals that for person #244, reducing their Activity Level while increasing their Willingness to

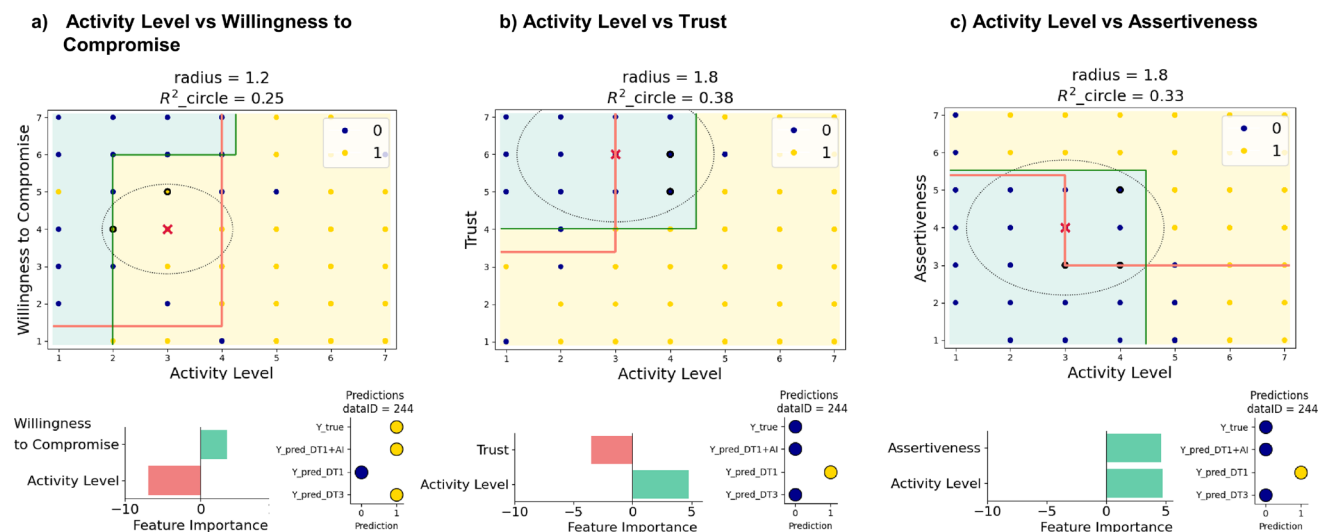


Fig. 2 Performance of BAPC on Real-World Data, depicting personal trait evaluations with an explanation generated by BAPC for person index 244 (red cross). In the subplots different pairs of features and the respective change of decision boundaries are depicted. Red/Green lines: decision boundaries before/after error amplification. Blue data points with black edges: modified data points after error amplification. Dashed circle: neighborhood for error amplification. The shifts in the decision boundaries (from red to green) show the importance of features and provide actionable guidance for the user on which features to increase or decrease. The importance of the features, and thus the explanations, are depicted in the Feature Importance bar plot. In the right lower corner the class prediction values for the different steps of the method are depicted: Y_{true} : true class label of person 244; $Y_{\text{pred_DT1}} + \text{AI}$: prediction of the surrogate model; $Y_{\text{pred_DT1}}$: final prediction of the first decision tree; $Y_{\text{pred_DT3}}$: final prediction of the second decision tree (color figure online)

Compromise would result in a predicted class label change from 0 to 1. This change signifies a transition from being deemed unsuitable for a leadership position to being considered suitable for such a role. The specific circle radius of 1.2 was selected because it maximized the coefficient of determination (R^2 -Score), which was calculated using all 36 overlapping data points within the circled neighborhood and yielded a value of 0.25.

Further analysis of the feature pair “Trust” and “Activity Level” (Figure 2b) suggests that decreasing “Trust” while increasing “Activity Level” would lead to a class label change from 1 to 0. This indicates a shift from being considered suitable for a leadership position to being classified as unsuitable. Consequently, it is recommended that these features remain unchanged for person #244. Alternatively, reversing the suggestion could strengthen their suitability for a leadership role. Here, a circle radius of 1.8 was chosen, as it resulted in a maximized R^2 -Score of 0.38 among the tested circle radii.

The optimized R^2 -Score values, along with their corresponding radii and the number of datapoints within the selected neighborhood for the R^2 -Score calculation, are shown in (Table 2).

3.4 Comparative analysis of explainability methods: BAPC vs. LIME

Notably, to the best of our knowledge, no existing method directly compares with our approach, particularly in explaining how changes to features influence the prediction of class labels.

To assess our method in a broader context, we compare it with LIME [19], a local surrogate technique that provides explanations for a single instance by using the notion of feature importance based on the model’s predictions in the instance’s neighborhood. We evaluate LIME on both datasets – synthetic and the application dataset – and compare its performance to our proposed method, BAPC with Error Amplification. The comparison focuses on two aspects: the suggested feature importance (explanation of the AI model) and the coefficient of determination (R^2 -Score), which measures how accurately the explanation reflects the predictions of the black-box model.

LIME uses a ridge regression model as a local surrogate to approximate the behavior of the black-box AI model around a specific instance. To measure how well the surrogate model represents the black-box model’s predictions, LIME evaluates the coefficient of determination (R^2 -Score), which acts as a measure of fidelity.

In LIME, two key parameters influence the class label prediction and the R^2 -Score:

- (1) *Kernel Width*: This parameter determines how much weight is given to perturbed samples based on their similarity to the original instance. Closer samples are weighted more heavily with a narrower kernel width, while a broader kernel considers a wider neighborhood.
- (2) *Number of Perturbed Samples*: This parameter specifies how many synthetic data points are generated around the original instance. These samples are used to fit the interpretable model, which approximates the black-box model’s local behavior.

These parameters are not predetermined and must be carefully tuned for each dataset. Finding the right balance between them is crucial to achieve a reliable approximation of the local surrogate model, leading to trustworthy predictions and explanations. [20, 29].

Table 2 Summary of performance results for various feature pairs of personal traits dataset with BAPC

Feature pair	R^2 -Score	Radius	Number of datapoints in neighborhood	Accuracy surrogate model
Willingness to Compromise vs Activity Level	0.25	1.2	36	0.994
Trust vs Activity Level	0.38	1.8	61	0.991
Assertiveness vs Activity Level	0.33	1.8	47	1.0

In this study, we compare the results of our proposed method, BAPC with Error Amplification, with LIME, to demonstrate its compatibility with existing methods while highlighting its improved performance and enhanced explainability.

For a fair comparison, we ensure the same number of data points is used to calculate the R^2 -Score in both methods. Specifically, LIME's perturbed samples are matched to the data points within the circular neighborhood of X_n used by BAPC. Additionally, we chose to set LIME's *kernel_width* equal to the *radius* in BAPC.

3.4.1 LIME on synthetic data - comparison to BAPC with Error Amplification

We apply LIME to the same synthetic dataset used with our method, as described in Sect. 3.1. The objective is to explain the black-box model's prediction for the specific data point of interest, $X_n = [2.5, 2.6]$. To ensure comparability, we also use a Random Forest Classifier as the AI model in this analysis.

For LIME, we select the parameters *kernel_width* and *number_perturbed_samples* as follows:

We match the number of perturbed samples in LIME to the number of neighborhood points in BAPC, which is $n_{\text{BAPC}} = 110$ samples, and set the *kernel_width* to the neighborhood radius of 1.4 used in BAPC. With this parameter configuration, LIME's surrogate model achieves an R^2 -Score of 0.09, indicating that it poorly approximates the behavior of the black-box model for the given data point and its neighborhood (Fig. 3b,c). In contrast, BAPC achieves an R^2 -Score of 0.44, providing a more reliable explanation of the AI model's behavior.

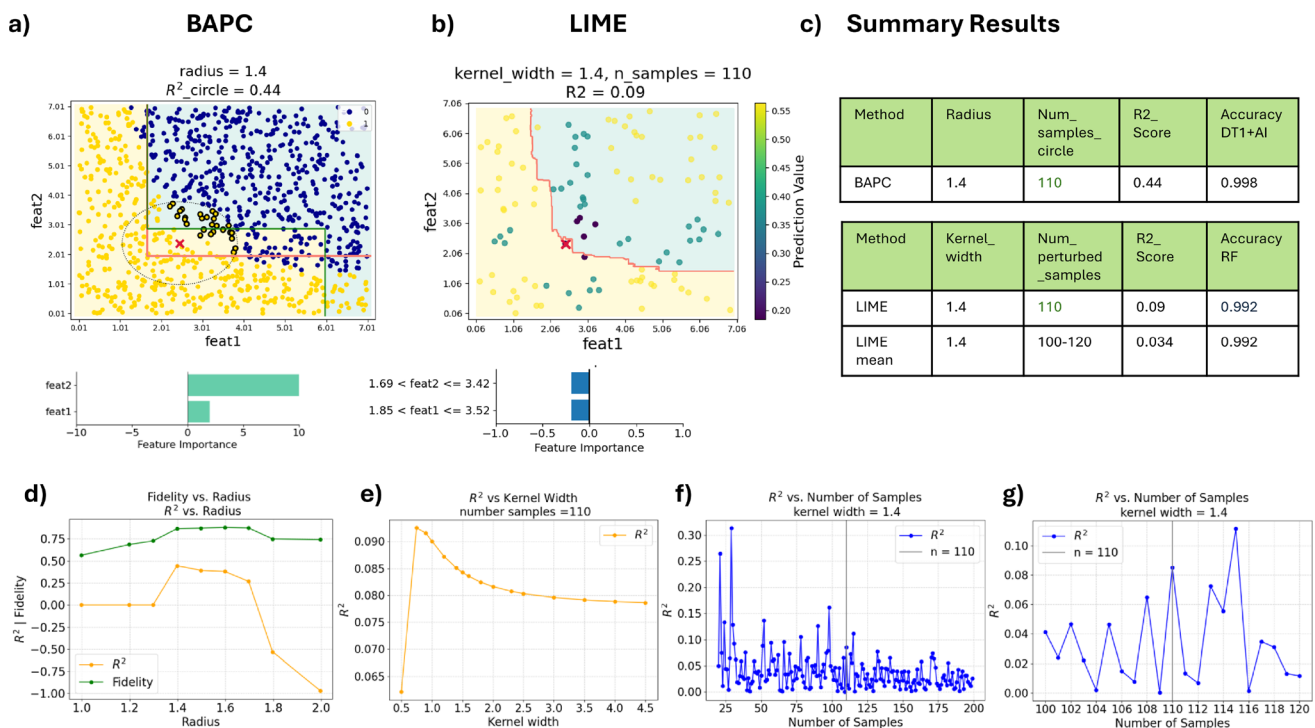


Fig. 3 Comparison of BAPC and LIME results on synthetic data. **a** Predictions with BAPC. Red/Green lines depict decision boundaries before/after error amplification. The shift in the decision boundary represents the explanation provided by BAPC, which is illustrated through feature importances. Data points are colored by true class labels. **b** Predictions with LIME, where points are colored by *Class 1* prediction values of LIME's surrogate model; opacity indicates sample weight, with higher opacity indicating higher weight. The decision boundary of the black-box model is shown with the red line. **c** Summary of parameter setup and resulting R^2 Score for BAPC and LIME. **d** Relationship between R^2 , Fidelity vs Radius for BAPC. **e** Relationship between R^2 and kernel width with LIME. **f** Relationship between R^2 and Number of perturbed samples with LIME. **g** Zoom-in on **f** for *num_perturbed_samples* = [100, 120] (color figure online)

The true label of the data point is *Class 1*, but LIME assigns a prediction probability of only 0.18 for *Class 1*. This demonstrates the limited predictive performance of the surrogate model with the chosen parameters.

Feature Importance. Analyzing the feature importances, LIME suggests that when feature 1 is in the range (1.85, 3.52), it negatively impacts the prediction probability of *Class 1*. This implies that moving feature 1 outside this range (note, in negative and positive, so decreasing or increasing the values of feature 1) would increase the probability of *Class 1*, potentially causing a class change (from *Class 0* to *Class 1*). A similar negative impact is observed for feature 2 within the range (1.69, 3.42), leading to the same conclusion about class prediction behavior as for feature 1.

However, caution is required when interpreting these results, given the low fidelity of LIME's local surrogate model, as indicated by the low R^2 -Score.

In contrast, BAPC identifies that increasing the values of either feature 1 or feature 2 causes a class label change from *Class 1* to *Class 0*. BAPC suggests that feature 2 has a greater impact on class label changes than feature 1, as even a small increase in its value leads to a class label shift, due to its proximity to the decision boundary. This interpretation aligns more closely with direct observations of the data and their corresponding class labels, offering a more plausible explanation in this context.

Exploration of R^2 . Additionally, we investigate the relationship between LIME's parameters and the R^2 -Score. We evaluate the R^2 -Score for different kernel widths (Fig. 3e) and numbers of perturbed samples (Fig. 3f, g) to explore how the R^2 -Score varies with different parameter combinations and whether a higher R^2 -Score can be achieved with alternative settings.

In general, the R^2 -Score shows high variability when changing the number of samples, even by a single sample. This suggests that relying on a single number of samples may not provide a reliable evaluation. Instead, we recommend using a range of sample sizes and calculating the mean R^2 -Score. We selected a sample size range of [100, 120], evaluated the R^2 -Score for each case, and found that the mean of these scores was 0.034. This is about one-third of the observed $R^2 = 0.09$ for the chosen $num_perturbed_samples = 110$. This emphasizes the importance of being cautious when selecting the number of samples, as it can significantly impact the R^2 -Score.

We analyze the relationship between the R^2 -Score and kernel width, finding the highest R^2 value of 0.0925 at a kernel width of 0.75, with 110 perturbed samples. Increasing the kernel width steadily lowers the R^2 -Score, stabilizing at around 0.079. The kernel width determines how neighborhood data points influence the analysis by assigning weights. At $kernel_width = 0.75$, the 6 closest data points are given the highest weight of 1, the next 35 points a weight of 0.41, and the furthest 69 points 0.17. Larger kernel widths, like $kernel_width = 2$, reduce weight differences, with weights distributed as (1, 0.88, 0.78), giving more equal importance across the neighborhood. For a fair comparison with our method, selecting an appropriate kernel width is crucial. It is important to balance the contribution of data points for the calculation of R^2 so that not only a small subset of data points dominates the evaluation (as in the case for smaller kernel widths), ensuring a fair representation of the neighborhood in the R^2 -Score assessment.

We compare the R^2 scores of BAPC and LIME using a one sided paired t-test across a sample population evaluated under varying radii (BAPC) ([1.35, 1.7]), corresponding kernel widths (LIME)([1.35, 1.7]), and associated sample sizes (see Appendix 7). The test examines the hypothesis that BAPC achieves higher R^2 scores compared to LIME. The resulting p-value of 0.0018 indicates a statistically significant difference, supporting the claim that BAPC consistently delivers better predictive performance than LIME in this example.

Evaluating the relationship in BAPC with R^2 and the Fidelity with the radius of the neighborhood (Fig. 3d), the results suggest, that the highest R^2 -Score is reached with a radius= 1.4.

Computation Time We compare the computation times of BAPC-DT and LIME on synthetic data across varying sample sizes. For BAPC-DT, the total evaluation time ranges from approximately 5.28 to 6.06 seconds for sample sizes of 250, 500, 1000, 1500, and 2000. This includes training two decision trees and one random forest classifier, as well as the relabeling process involved in the error amplification step. Among the individual steps, error amplification is the most time-consuming, consistently taking around 4 seconds across all sample

sizes. It is important to note that the current implementation has not been optimized for speed, and there is significant potential for improvement through more efficient computational methods. In contrast, LIME runs notably faster, with computation times between 0.16 and 0.18 seconds across the same sample sizes. Although BAPC-DT demands more computation, its runtime remains within a reasonable range. Nevertheless, LIME clearly holds an advantage in speed in this particular setting. A full table detailing computation times for each sample size can be found in Appendix 3.

3.4.2 LIME on application data - comparison to BAPC with Error Amplification

We apply LIME to the personal traits dataset introduced in Sect. 3.2 and compare the results for the feature pair *Assertiveness* vs *Activity Level* with those obtained using BAPC with Error Amplification. A similar analysis for *Trust* vs *Activity Level* is provided in Appendix 6. Note, that this dataset consists of discrete values ranging from [1, 7].

Feature Importance. When comparing the feature explanations, the predicted feature importance for *Activity Level* appears qualitatively similar across both methods. LIME indicates that when *Activity Level* is within the range (0, 4), it negatively impacts the probability of predicting *Class 1*. This suggests that values within this range make the model less likely to classify the instance as belonging to *Class 1*. On the other hand, when *Activity Level* exceeds 4, the probability of predicting *Class 1* increases.

Similarly, BAPC suggests that *Activity Level* has a positive influence on the class label. As *Activity Level* increases, the class label shifts from 0 to 1, reflecting the same qualitative trend as observed with LIME.

Regarding the feature *Assertiveness*, LIME suggests that when *Assertiveness* is equal to 4, it has only a minimal negative impact on the probability of predicting *Class 1*. Thus, values of *Assertiveness* both smaller and larger than 4 appear to have little effect on the prediction for *Class 1*, as indicated by the prediction values remaining in the blue range along the *Assertiveness* axis (Fig. 4b).

In contrast, BAPC suggests that *Assertiveness* plays a similarly important role as *Activity Level*, with an equally positive impact. Specifically, increasing the *Assertiveness* value leads to a shift in the predicted class label from *Class 0* to *Class 1*. This relationship is also evident upon visually examining the distribution of the actual data and their corresponding class labels, reinforcing its validity as a plausible explanation (Fig. 4a).

Exploring R^2 . Since BAPC achieves the highest R^2 -Score with a radius of 1.8 (Fig. 4d), we set LIME's *kernel_width* to 1.8 for a fair comparison. In BAPC, this radius includes 47 data points for the R^2 -Score calculation, so we set the number of perturbed samples in LIME to 47 as well.

Evaluating the R^2 -Score, LIME achieves $R^2_{LIME} = 0.39$, which is comparable to BAPC's $R^2_{BAPC} = 0.33$ (Fig. 4c). However, the R^2 -Score for LIME shows significant variability depending on the number of perturbed samples (Fig. 4f). To account for this, we calculate the mean R^2 -Score over a range of perturbed samples between [37, 57], resulting in $R^2_{mean} = 0.36$.

4 Discussion

Our proposed method provides a systematic approach to explaining and, if necessary, modifying the predicted class of a specific data point. Unlike many explainability techniques that only aim to interpret model decisions, our approach goes a step further by offering assistance for decision making. This is achieved through the use of feature importance (4) (see discussion in [6]) and by analyzing how the decision boundaries of the base model change.

If a given point is originally classified as *Class 1*, but we seek a transition to *Class 0*, our method identifies which features should be intensified or reduced to drive this class change. This enables a deeper understanding of the decision-making process by not only explaining why a prediction was made but also indicating how targeted feature adjustments can influence classification outcomes.

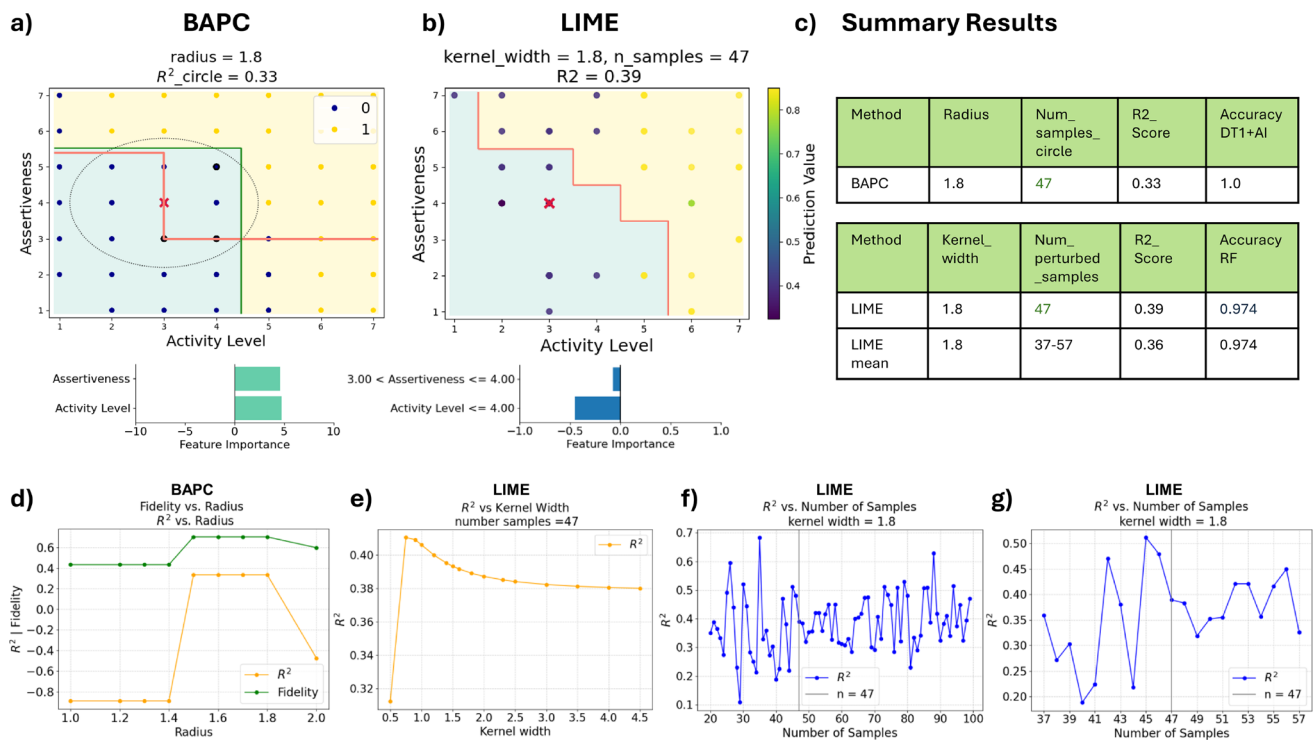


Fig. 4 Comparison of BAPC and LIME results on personal traits dataset for features *Assertiveness* vs. *Activity Level*. **a** Predictions with BAPC. Red/Green lines depict the decision boundaries before/after error amplification. The shift in the decision boundary represents the explanation provided by BAPC, which is illustrated through feature importances. Data points colored by true class labels. **b** LIME predictions colored by Class 1 values from the surrogate model, with opacity indicating sample weight, with higher opacity indicating higher weight. The decision boundary of the black-box model is shown with the red line. **c** Summary of parameter setup and resulting R^2 -Score and accuracies of the surrogate model in BAPC (DT1+AI) and random forest accuracy in LIME. **d** Relationship between R^2 , Fidelity vs. Radius for BAPC. **e** Relationship between R^2 and Kernel width with LIME. **f** Relationship between R^2 and number of perturbed samples with LIME. **g** Zoom-in on **f** for $\text{num_perturbed_samples} = [37, 57]$ (color figure online)

Since our method introduces a novel concept via the break-up into base model and correcting AI model, direct comparisons with existing techniques requires careful consideration. We selected LIME as a baseline for comparison because it similarly focuses on explaining model predictions at an instance level by considering a modification of the data set (sampling(LIME) vs. relabeling(EA)). LIME provides insights into feature contributions for a specific data point, offering an interpretation of why a classification was made. However, while both LIME and our method provide explanations for predictions, our approach goes further by also suggesting concrete changes to influence classification results. This distinction highlights the added value of our approach in scenarios where not only interpretability but also guided decision adjustments are required [7].

To quantitatively assess the quality of explanations, we evaluate the **fidelity score** (7), which measures how accurately an explanation reflects the predictions of the black-box model. Additionally, we report the R^2 **score** (8) for our method, as it is also used in LIME as a fidelity metric, allowing for a direct numerical comparison between the approaches. Furthermore, we assess the **accuracy** of the surrogate model and we compare this accuracy to the performance of the Random Forest surrogate model used in LIME.

A direct comparison between our method and LIME on simulated data (Sect. 3.3.1) shows that BAPC outperforms LIME, achieving an R^2 score of 0.44 compared to 0.09 for LIME (Fig. 3c). This suggests that the predictions made by LIME's surrogate model—and consequently its explanations—may not be highly reliable in this case.

Additionally, we observe that the R^2 score in LIME is highly sensitive to the number of perturbed samples (Fig. 3f,g). When evaluating the R^2 score for 100 – 120 perturbed samples, we obtain a $mean(R^2)$ score of 0.034, indicating strong variability depending on the sample size. This suggests that relying on a single R^2 score for a fixed number of perturbed samples can be misleading, and careful consideration is required when interpreting LIME’s fidelity metrics. To provide a comparison, our method enables the analysis of R^2 variability based on the chosen **radius**, which is the only parameter that needs to be set. The radius determines how many neighborhood samples are considered for error amplification. We observe that R^2 remains relatively stable within specific radius intervals (Fig. 3d), showing no significant fluctuations, except for discrete jumps when transitioning ranges between different radii. The consistent performance under different radius settings reflects the robustness of BAPC-DT’s design.

For our domain specific personal traits dataset, we conducted a similar comparison (Sect. 3.3.2). The results show that both methods achieve comparable model reliability, as evaluated using the R^2 score: BAPC achieves 0.33, while LIME attains with the $mean(R^2) = 0.36$ (Fig. 4c).

However generally, both methods exhibit relatively low R^2 scores, suggesting that the surrogate models may not be highly reliable in this particular dataset. A key characteristic of this dataset is its small size and low resolution, as it contains only integer values between [1, 7], which limits its predictive power and the overall explainability of the models, as reflected in the R^2 scores.

Despite these results, we chose to include the R^2 score because it is the standard fidelity measure used by LIME. Aligning with this choice allows for a consistent and comparable evaluation between BAPC and LIME. Nevertheless, our results indicate that R^2 may not be the most informative metric in this setting. To address this, we also report fidelity scores for BAPC-DT, which better capture the local approximation quality of the surrogate models. In the synthetic data scenario, fidelity reaches 0.87 at a radius of 1.6 (Fig. 3d), and for real-world data, 0.7 at a radius of 1.5 (Fig. 4d) - both suggesting good local fidelity. Interestingly, when comparing fidelity values and R^2 scores, we observe that both metrics tend to peak within the same range of radii - showing that higher R^2 scores coincide with higher fidelity scores.

Therefore, we recommend using fidelity as a more suitable measure for evaluating surrogate performance in such cases.

4.1 Interpretability of results for the personality traits dataset

Beyond quantitative comparisons, we also analyze the interpretability of the results provided by both methods. For a specific data point of interest, which was initially predicted by the base model as *Class 0* (which can be interpreted as “not suitable for a leadership position”), our method provides the following insight:

- A simultaneous increase in the features *Activity Level* and *Assertiveness* would lead to a class transition to *Class 1*. (Fig. 4a)

In contrast, LIME provides the following explanation:

- The feature called *Activity Level* has a stronger negative influence on the final prediction of *Class 0* for all values below 4. (Fig. 4b) This means that data points with an *Activity Level* ≤ 4 contribute to the model’s decision to classify the instance as *Class 0*. One could interpret this as implying that an increase in *Activity Level* above 4 would lead to a *Class 1* prediction, which aligns with our method’s results.

However, there is a notable difference in interpretability between the two methods regarding the personality feature called *Assertiveness*:

- LIME suggests that *Assertiveness* (for values in the range [3,4]) has only a minimal negative impact on the *Class 0* prediction for this instance. This implies that *Assertiveness* does not significantly influence a change in classification.

- In contrast, our method suggests that both *Activity Level* and *Assertiveness* contribute equally to a potential class change.

Upon inspecting the raw training data, we observe that increasing *Assertiveness* does indeed lead to a class transition, reinforcing the validity of our method's explanation. This suggests that BAPC provides a more actionable and realistic interpretation in this case.

4.2 Overfitting and model complexity considerations

An important advantage of our method in this setting is its use of a simple decision tree as the base model, which helps mitigate overfitting on low-resolution data. This allows our method to capture general trends and provide explanations that reflect the broader patterns within the dataset. In contrast, LIME relies on Ridge regression as its surrogate model making it more prone to overfitting in this low-resolution scenario.

This is where our hybrid model architecture proves beneficial, as it effectively balances interpretability and predictive refinement. Additionally, our method handles discrete data well, making it particularly suited for scenarios where feature values are limited to a predefined set of categories. This difference demonstrates a key strength of our approach, as it enables stable, interpretable insights while avoiding the pitfalls of excessive model complexity in cases where data resolution is limited.

4.3 Definition of the neighborhood: BAPC vs. LIME

A crucial aspect of both methods is the definition of the neighborhood around the instance being explained ('locality problem'). In BAPC, this is controlled by the radius parameter, which defines how many points within a given radius are considered for error amplification. In LIME, neighborhood definition is influenced by the kernel width parameter, which determines how much weight is given to perturbed samples based on their distance from the original data point.

Given that our personality traits dataset consists of only discrete values, we can evaluate the R^2 score across different radius values and select the radius with the highest R^2 score. This provides a structured approach for parameter selection in BAPC (cf. Sect. 2.4 in [1]).

In contrast, choosing an optimal kernel width in LIME is less straightforward because R^2 scores depend on both the number of perturbed samples and the kernel width parameter. In our study, we aligned LIME's parameters with our method's settings to ensure comparability. However, finding the best LIME parameters defining the neighborhood remains ambiguous, as also noted in other studies ([20, 29]).

This sensitivity in parameter selection shows a potential limitation of LIME, particularly in datasets with low resolution, such as the one used in our study.

5 Conclusion

We introduced EA as an improved local surrogate method coping with the problem of finding explanations of predictions on small discrete data sets for which overfitting is an eminent issue for high performing machine learning methods. The method relies on the BAPC concept of using a hybrid decomposition of the predicting model to yield explanations of the base model's parameters. The specific choice of decision trees as the base model allows changes in decision boundaries to produce a map-like overview in the neighborhood of the instance. This supports decision-making based on feature importance scores, which are also produced by our method.

The advantage over LIME in only having to use a single parameter (radius of the neighborhood) makes EA practical for guiding decision makers due to the reduced variability of the fidelity score in comparison to the highly dispersive R^2 -metric of LIME.

Appendix

In the Appendix, we provide the following supplementary materials:

- Figure 5: Ablation Study - BAPC without Error Amplification.
- Figure 6.: Comparison of BAPC and LIME results on personal traits data set with features “Trust” vs “Activity Level”
- Figure 7: Statistical evaluation for comparing R^2 Scores for BAPC vs LIME using a paired t-test.
- Table 3: Computation time comparison of BAPC-EA and LIME across sample sizes for the synthetic dataset.

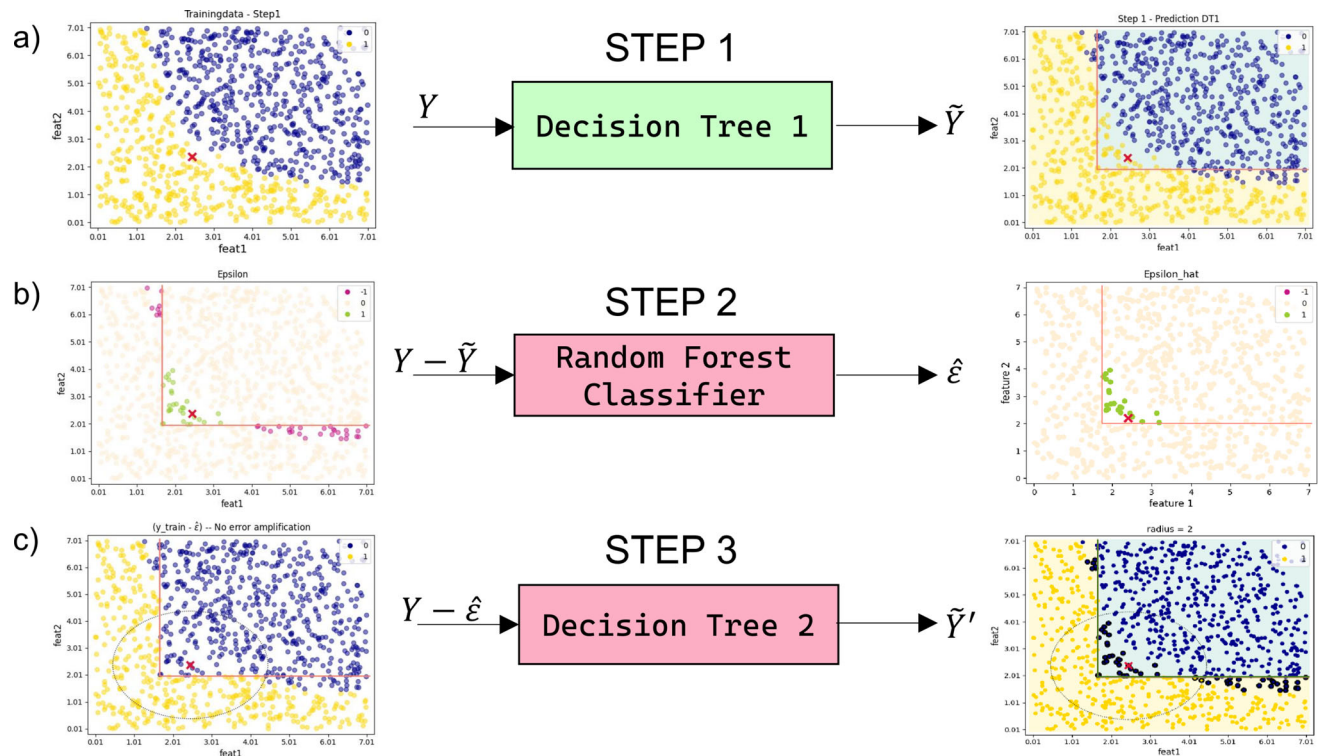


Fig. 5 Ablation Study - Schematic Overview of BAPC without Error Amplification. Discrete data points are shown for two features belonging to two categories $\{0\text{--blue}, 1\text{--yellow}\}$. The data point of interest X_n is depicted with the red cross. The red/green lines correspond to the decision boundaries before/after error amplification. **a** Step 1: Train first decision tree model and get predictions $\tilde{Y}$. **b** Step 2: Infer the error of the predictions of the first decision tree $Y - \tilde{Y}$ and predict the error with an AI model $\hat{\epsilon}$ in a circular neighborhood around X_n . **c** Step 3: Input data are the modified dataset $Y - \hat{\epsilon}$. A second decision tree is trained on this relabeled data, resulting in the same decision boundaries as the first decision tree, which means no shift in the decision boundaries occurs (color figure online)

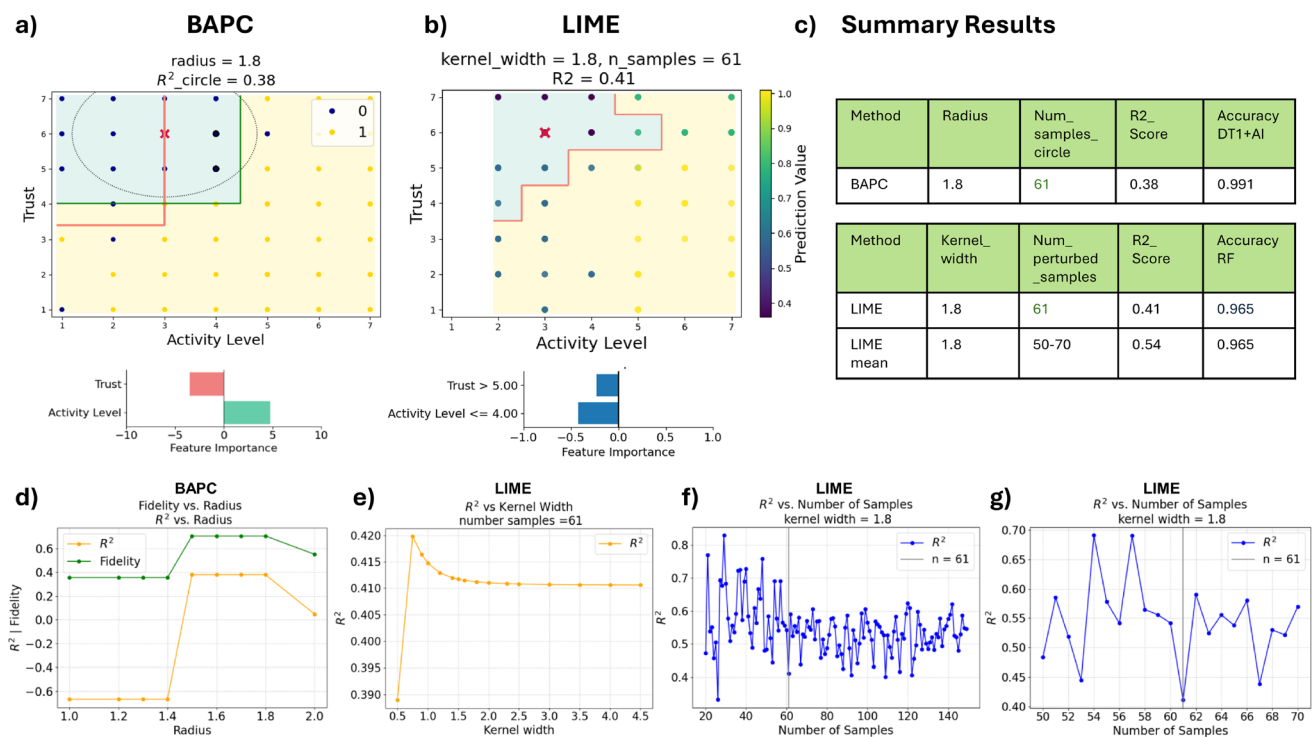


Fig. 6 Comparison of BAPC and LIME results on personal traits data set with features “Trust” vs “Activity Level”. **a** Predictions with BAPC. Red/Green lines depict the decision boundaries before/after error amplification. The shift in the decision boundary represents the explanation provided by BAPC, which is illustrated through feature importances. The colors of the data points correspond to the true class labels. **b** Predictions with LIME. The data points are colored according to the prediction values for Class 1 from LIME’s surrogate model. The opacity of the data points reflects their weight, with higher opacity indicating higher weight. The decision boundary of the black-box model is shown with the red line. **c** Summary of parameter setup and resulting R^2 Score for BAPC and LIME. **d** Evaluation of the relationship Radius vs. Fidelity and Radius vs. R^2 for BAPC. **e** Relationship between R^2 and kernel width with LIME. **f** Relationship between R^2 and number of perturbed samples with LIME. **g** Extract of **f** showing a zoom for $\text{num_perturbed_samples} = [100, 120]$ (color figure online)

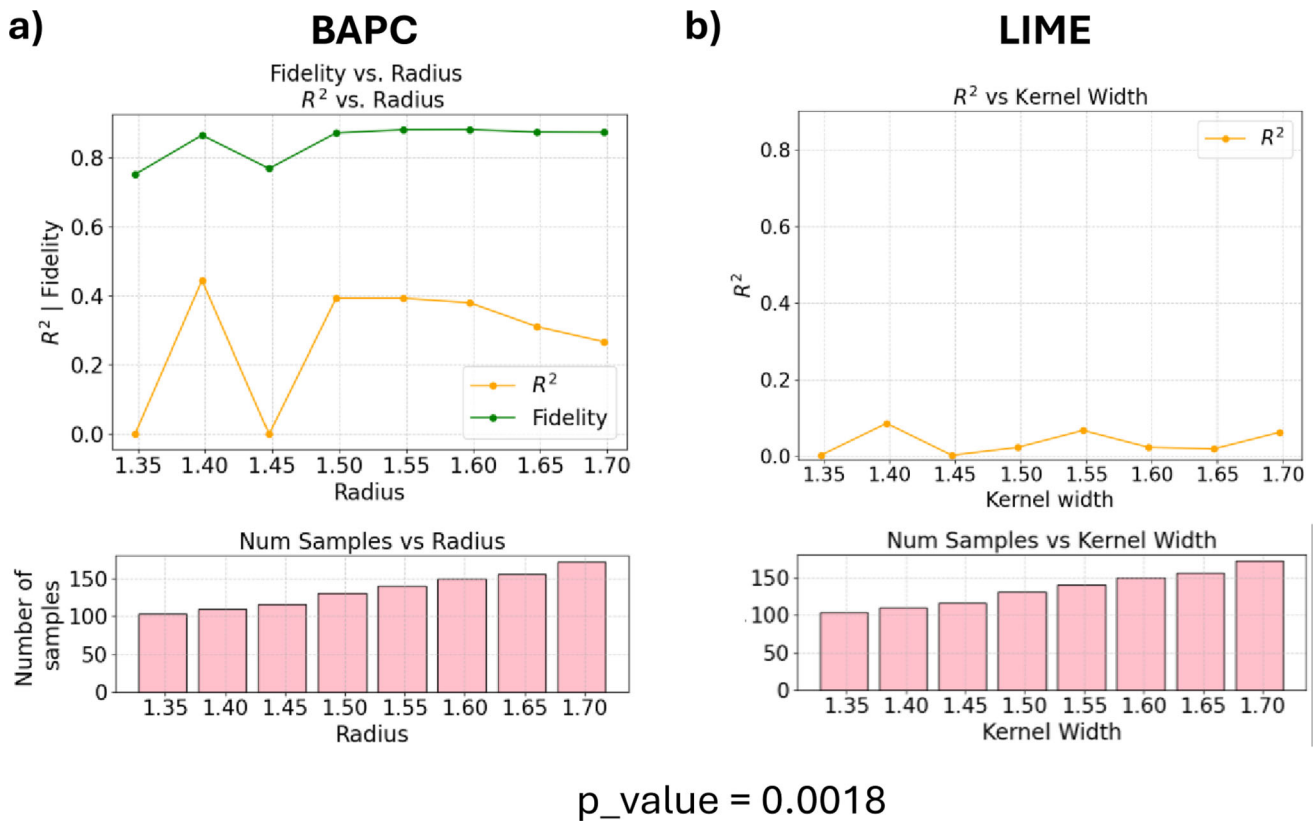


Fig. 7 Comparison of R^2 Scores for BAPC vs LIME using a paired t-test for different radii / kernel widths and respective sample sizes results in a p-value of 0.0018. **a** Top: R^2 Scores plotted versus Radius evaluated with BAPC on simulated dataset. Below: the plot depicts the number of samples in the neighborhood for the given radius. **b** Top: R^2 scores plotted against kernel width, evaluated using LIME on the simulated dataset. Bottom: The number of perturbed samples used for each kernel width. These were selected to match the number of samples in BAPC's local neighborhood to ensure a fair comparison

Table 3 Computation time comparison of BAPC-EA (by step) and LIME across sample sizes for the synthetic dataset

Sample size	BAPC step 1 (s)	BAPC step 2 (s)	BAPC EA (s)	BAPC step 3 (s)	Total BAPC-DT (s)	LIME (s)
250	0.54	0.04	4.05	0.64	5.28	0.16
500	0.56	0.05	4.20	0.72	5.52	0.13
1000	0.60	0.07	4.07	0.85	5.59	0.16
1500	0.62	0.09	4.19	0.92	5.82	0.17
2000	0.64	0.11	4.18	1.13	6.06	0.18

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Author contributions E.D.: Conceptualization of the method, Formulation of mathematical model, Algorithmic implementation of the method, Preparation of figures, Writing – original draft, review and editing; F.S.: Conceptualization of the method, Formulation of mathematical model, Writing – original draft, review and editing.

Data availability This work is implemented in Python (v3.10), and the code is available on GitHub [30]. Simulated data (Sect. 3.1, Figure 3) is available on GitHub [31]. The application data set (Sect. 3.2, Figs. 2, 4, 6) is available in an anonymized version on GitHub [31].

Declarations

Conflict of interests The authors declare that they have no Conflict of interests.

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